

Day 26 - OSPF (pt 1)

Sunday, September 6, 2026 10:46 AM

OSPF Information

Open Shortest Path First (OSPF) is a dynamic routing protocol that is widely used in modern networks. It uses the Shortest Path First (SPF) algorithm (also called Dijkstra's algorithm)

- **OSPF** has three versions:
 - **OSPFv1**: Old and not in use anymore.
 - **OSPFv2**: Used for **IPv4**
 - **OSPFv3**: Used for **IPv6** (can also be used for **IPv4**, but usually **v2** is used)

Routers store information about the network in LSAs (Link State Advertisements) which are organized in LSDBs (Link State Databases)

Routers will **flood** LSAs until all routers in the OSPF area develop the same map of the network. (until all routers have the same LSDB)

LSA Flooding Process

- When **OSPF** is activated on an interface, the router creates an **LSA** to tell its neighbors about the network on that interface.
- The **LSA** is **flooded** throughout the network until it has been received by all routers.
- Eventually, all routers will have the same **LSDB** and each router will use the **SPF** algorithm to calculate the best routes

In OSPF, the three main steps to sharing LSAs and determining the best route are:

1. Become neighbors with other routers
2. Exchange **LSAs** with neighbor routers
3. Calculate the best routes

OSPF Areas

OSPF uses areas to divide up the network.

Small networks can use a **single-area** setup without any negative effects.

However, large networks can suffer from many negative effects if a **single-area** setup is used.

The effects of **single-area** on large networks

- The **SPF** algorithm takes more time to calculate routes
- The **SPF** algorithm requires exponentially more processing power on the routers
- The larger **LSDB** takes up more memory on the routers
- Any small change in the network causes every router to flood **LSAs** and run the **SPF** algorithm again

OSPF Terminology

- Area: An area is a set of routers and links that share the same LSDB.
- Backbone Area (area 0): A special area that all other areas in the network must connect to.
- Internal Routers: Routers with all interfaces in the same area.
- Area Border Routers (ABR): Routers with interfaces in multiple areas.
- Autonomous System Boundary Router (ASBR): OSPF router that connects the OSPF network to an external router.
- Backbone Routers: Routers connected to the backbone area.
- Intra-area Route: A route to a destination inside the same OSPF area.
- Interarea Route: A route to a destination in a different OSPF area.

OSPF Area Rules

- **OSPF areas** should be **contiguous**.
 - (Being **non-contiguous** would mean having a **area 1** be split up into two different places, for example)
- All **OSPF areas** must have at least one **ABR** connected to the **backbone area**.
- **OSPF** interfaces in the same subnet must be in the same **area**.

ABRs Maximum Area Connections

- ABRs maintain a separate **LSDB** for each **area** they are connected to.
- It is recommended that you connect an **ABR** to a maximum of **2 areas**.
 - Connecting an **ABR** to **3+** areas can overburden the router.

OSPF Autonomous System Boundary Router (ASBR)

The Autonomous System Boundary Router (ASBR) is an OSPF router that connects the OSPF network to an external router.

- It is assigned when a router has a **default route** and the **default-information originate** command is used on the router.

Basic OSPF Configuration

OSPF initialization is done through the **router ospf PROCESS-ID area AREA-ID** command.

The OSPF **process ID** is locally significant on the router itself. Routers with different **process IDs** can become OSPF neighbors, they do not need to have matching **process IDs**.

- **Network command:**
 - The **network** command in **OSPF** requires you to specify the **area**.
 - Ex. **network IP-ADDRESS WILDCARD-MASK area AREA**
 - Also, the **network** command tells the router to:
 - Look for interfaces with an IP address that is in the specified range and activate **OSPF** on them in the specified **area**.
 - The router will then try to become **OSPF** neighbors with other **OSPF** routers.
- **Passive Interface command:**
 - The **passive-interface INTERFACE-ID** command tells the router to stop sending **OSPF Hello** messages out of the specified interface. It is used on interfaces that don't have any **OSPF** neighbors.
 - However, the router will still continue to send **LSAs** informing its neighbors about the subnet configured on that interface.
- **Default Information command:**
 - The **default-information originate** command tells the router to advertise its **default route** to the other protocol neighbors.

OSPF Router ID

The OSPF router ID is selected through **3** methods, with priority being from top to bottom

1. Manual Configuration (via **router-id A.B.C.D**)
2. Highest IP address on a **loopback** interface
3. Highest IP address on a **physical** interface

Watch out: Router-ID functionality

- If **OSPF** has already been started on the device (through **router ospf PROCESS-ID area AREA-ID**) and a **Router-ID** has been assigned either manually or automatically, then the **Router-ID** will stay the same, and will not change until the **OSPF** process is restarted.
- Even if **router-id A.B.C.D** is used to change the **Router-ID**, the change will not take effect until an **OSPF** restart is performed.